

Anomaly Detection Workflow		Human Capabilities and Execution		LLM Capabilities and Empowerment	
	(a) Anomaly Detection Tool-dependent Task Advanced anomaly detection algorithms (statistical, machine learning, deep learning, etc.)		Telemetry Data Analysts Tool Operation and Professional Skills Debug and deploy anomaly detection algorithms, set thresholds and monitor system data		LLM Function Calling and Tool Integration Function Call, MCP (Model Context Protocol)
	(b) Work Condition Recognition Logical Reasoning Task Well-defined logic rule engines, decision trees, etc.		Space Mission Operators Logical Reasoning and Judgment Monitor telemetry and calculate relevant indicators to determine the system's work condition		LLM Reasoning and Chained Thinking Prompt Engineering, CoT (Chain-of-Thought), Step-wise Prompt, Zero Shot
	(c) Anomaly Localization Learning-and-approximation Task Historical anomaly cases and anomaly monitoring signals, etc.		Senior Subsystem Experts Experience Accumulation and Pattern Learning Continuous research and summarize to accumulate experience, analyze data to identify anomaly type		LLM Learning and Approximation from Data Fine-Tuning, LoRA (Low-Rank Adaptation), GRPO (Group Relative Policy Optimization)
	(d) Maintenance Decision-making Knowledge-intensive Task Maintenance manuals, technical documents and operating procedures, etc.		On-orbit Response Teams Search, Summarization and Decision-making Consult technical documents, maintenance manuals, relevant procedures to formulate maintenance plans.		LLM Knowledge Base Querying and Synthesis RAG (Retrieval-Augmented Generation), MCP (Model Context Protocol), Knowledge Base

能力：学习
 操作：调试